Project Report: 4-Axis Custom Robotic Manipulator

This document provides a comprehensive technical overview of the development, hardware architecture, and software logic of a custom-designed 4-axis robotic arm. The system features a joystick-controlled interface and a specialized end-effector design.

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1. Project Overview

The objective of this project was to design and build a functional robotic arm capable of multi-axis movement using cost-effective components and custom 3D-printed parts. The arm utilizes four degrees of freedom (DoF) for positioning, with a fifth actuator dedicated to the gripper mechanism.

- **Controller:** Arduino Uno / Mega
- **Input:** Dual Analog Joysticks
- **Actuators:** 5 Servo Motors
- **Design Software:** Autodesk Fusion 360

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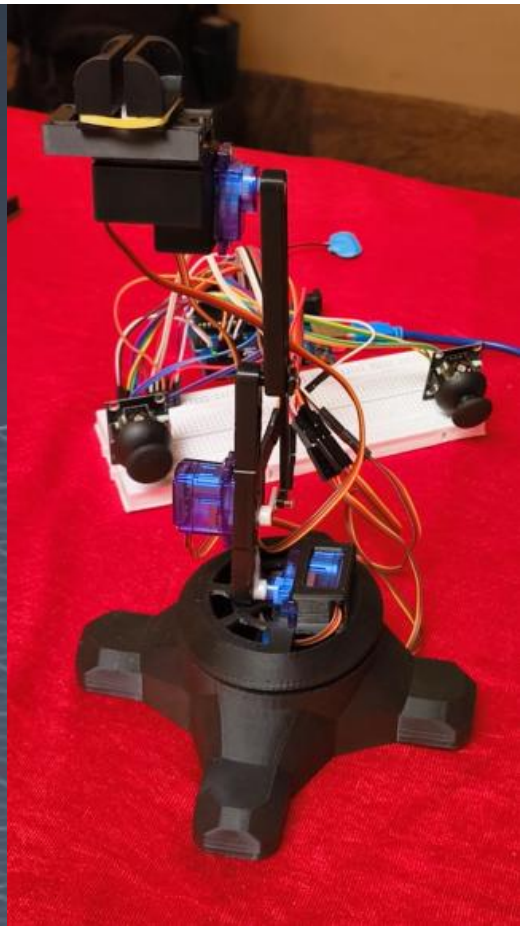
2. Mechanical Design & 3D Modeling

The physical structure of the manipulator was designed from scratch using **Fusion 360**. Key design considerations included:

- **Linkage System:** The arm comprises a rotating base, a shoulder joint, an elbow joint, and a wrist joint.
- **End Effector (Gripper):** A unique mechanical design where a single servo motor provides the opening force, while **elastic rubber bands** are integrated to facilitate the retraction/closing tension. This "passive return" mechanism ensures a constant gripping pressure without stalling the motor.
- **Structural Integrity:** The parts were optimized for 3D printing, ensuring enough clearance for servo fitment and wire routing.



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3. Hardware Components

3.1 Actuators

While the current prototype utilizes **SG90 Micro Servos**, there is a known limitation regarding torque and gear durability under load.

Note on Upgrading: For improved precision and higher load-bearing capacity, it is highly recommended to upgrade to **MG90** or **MG90S** servos. These offer metal gears and higher stall torque compared to the nylon gears of the SG90.

3.2 Electronics & Control

- **Analog Joysticks:** Two modules are used to provide four axes of analog input (X1, Y1, X2, Y2).
- **Breadboard Power Distribution:** Due to the high current draw of five servos, a central power rail on a breadboard is utilized.
- **Arduino Interface:** The controller maps analog voltage changes (0V to 5V) from the joysticks to PWM (Pulse Width Modulation) signals for the servos.

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4. Wiring Diagram and Pin Mapping

Based on the system architecture, the following pin configuration is established:

Component	Arduino Pin	Function
Servo 1	**Pin 3**	Base Rotation (PWM)
Servo 2	**Pin 5**	Shoulder Joint (PWM)
Servo 3	**Pin 6**	Elbow Joint (PWM)
Servo 4	**Pin 9**	Wrist/Link (PWM)
Servo 5	**Pin 10**	Gripper (PWM)
Joystick 1 VRx	**A0**	Base Control
Joystick 1 VRy	**A1**	Shoulder Control
Joystick 2 VRx	**A2**	Elbow Control
Joystick 2 VRy	**A3**	Wrist Control
Joystick Button	**Pin 2**	Gripper Toggle (Pull-up)

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5. Software Logic

The Arduino firmware follows a continuous loop execution:

1. **Analog Read:** The system samples the voltage from the four joystick axes.
2. **Mapping:** The raw values (to) are mapped to angles (to) or used as incremental step values to maintain the arm's position when the joystick returns to neutral.
3. **PWM Generation:** The Servo.h library sends the calculated position to the respective signal pins.
4. **Gripper Logic:** The end effector is often mapped to a joystick button or a specific threshold on the analog axis to trigger the opening sequence against the rubber band tension.

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6. Conclusion and Future Scope

The current iteration successfully demonstrates the feasibility of a custom-designed 4-axis arm using Fusion 360 and Arduino. Future improvements will focus on:

- Transitioning to **MG90S servos** for better payload handling.
- Implementing an external 5V power supply to prevent Arduino resets during simultaneous motor movement.
- Integrating Inverse Kinematics (IK) for more fluid, coordinate-based movement.